

Not Finished, But Abandoned

by

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ACKNOWLEDGMENTS

It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....

Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.

— DAVID C. MAKINSON (1965)

Above is the famous "preface paradox," which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

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ABSTRACT

FIXME: basically a placeholder; do not believe

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,
2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

1 INTRODUCTION

1.1 What the heck is a plasma

1.1.1 Plasmas in the universe

1.1.2 How do plasmas affect us on earth?

1.1.3 Where does magnetic reconnection play into this whole shabang?

1.2 Magnetic Reconnection

1.2.1 The Generalized Ohm's Law

1.2.1.1 How do we derive this?

1.2.1.2 A breakdown of the terms

1.2.2 Regimes of Reconnection

1.2.2.1 Reconnection scaling parameters

i.e. why do we use S and L/d_i to characterize reconnection regimes?

Figure 1.1: Diagram of reconnection regimes.

Figure 1.2: Diagram of Sweet-Parker reconnection.

Figure 1.3: Diagram of 2d hall reconnection

1.2.2.2 Where do various systems line up?

1.2.2.3 What are we interested in?

1.2.3 Sweet-Parker Reconnection

1.2.3.1 A quick historical detour

1.2.3.2 What are the assumptions that go into Sweet-Parker?

1.2.3.3 Sweet-Parker Reconnection Rate Derivation

1.2.3.4 Application to Real Systems

1.2.3.4.1 What are some example parameters where reconnection occurs?

1.2.3.4.2 What time scales does Sweet-Parker predict?

1.2.3.4.3 What did we miss?

1.2.4 Hall Reconnection

1.2.4.1 Why should the Hall term appear?

1.2.4.1.1 What is the relevant physics that sets when the hall term matters

1.2.4.1.2 In what kind of plasmas does the hall term appear and when does it not?

Figure 1.4: Flux tube changing size during reconnection causing pressure anisotropy build up.

Figure 1.5: Simulated kinetic reconnection with changing guide field.

1.2.5 Kinetic Reconnection

1.2.5.1 What makes kinetic reconnection “kinetic”?

1.2.5.2 Where does Pressure Anisotropy come from?

1.3 Experimental Setup

2 PRESSURE WAVE

Figure 2.1: Greess et al. 2022 paper showing simulations with pressure wave.

3 MULTI-TIP LANGMUIR PROBE

4 PRESSURE ANISOTROPY PROBE

5 CONCLUSION

DISCARD THIS PAGE

COLOPHON

This template uses Gyre Pagella by default. (I used Arno Pro in my dissertation.)

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